



NAUVAL ZULFIKAR

Senior Data Scientist · Operations Research & Simulation Bandung, Indonesia ·
zulfikar.nauval1998@gmail.com · +62 821-1704-3831 [LinkedIn](#) · [GitHub](#) · [Portfolio](#) **Eligible to work
and study in the UK (Graduate Visa active)**

PROFILE

Senior Data Scientist with 5+ years of applied ML experience in licensed banking, UK start-ups, regional government, and healthcare non-profit settings. **MSc Business Analytics, Aston University, 1:1 First Class** (1st Winner, Aston Enterprise Scholarship 2023). MSc dissertation supervised by Dr. Ammar Al-Bazi on blockchain-integrated LLM analysis for supply-chain customer reviews; methodological foundation for the proposed PhD project *A Digital Twin Framework for Behaviourally-Responsive Integrated Passenger-Freight Transport with Crowd-Shipping* (Aston, p196342).

FORMAL EDUCATION

Aston University, Birmingham, UK (*MSc Business Analytics, 1:1 First Class · EQF Level 7 · Sep 2023 – Sep 2024*)

- **1st Winner, Aston Enterprise Scholarship 2023** (full-tuition competitive award)
- **Aston Pitching Contest 2024** runner-up (AI-powered supply chain proposal)
- **Distinctions:** Decision Models, Effective Management Consultancy, Software Analytics
- **Dissertation:** *Enhancing Supply Chain Information Sharing Systems Through Blockchain: Integrated Customer Reviews LLM Analysis: A Smart Contract Approach* (supervised by **Dr. Ammar Al-Bazi**, Aston Business School)

Ritsumeikan Asia Pacific University, Beppu, Japan (*BBA Marketing · EQF Level 6 · Oct 2016 – Sep 2019*)

- APU 50% Tuition Reduction Scholarship + JASSO Scholarship (Japan Student Services Organisation)
- Distinctions: Business Data Analysis, Consumer Behaviour, Marketing Research

INFORMAL EDUCATION

Indonesia AI (*NLP Engineer Training · EQF Level 5 · May – Nov 2023*) — Contributed to the *2024 Indonesia Election Aspect-Based Sentiment Analyser* project. Distinctions: Python Programming, Text Summarisation, Voice Recognition.

Qarir Generator (Dubai, UAE) (*Data Science Global Accelerator · EQF Level 5 · Apr – Jun 2023*) — Led sentiment-analysis development for the *Pfizer Public Sentiment Analysis* project. Distinctions: Python, SQL, Pandas/NumPy.

RESEARCH PROJECTS (PHD APPLICATION PORTFOLIO, MAY 2026)

Mini Digital Twin for Integrated Passenger-Freight Transport (*Solo builder, May 2026*)

- Three-layer Digital Twin prototype: FastAPI mock-IoT, mesa-based ABM + simple DES, Streamlit dashboard with live folium map + plotly KPIs + interactive disruption injection (rain, breakdown, surge). Smoke-tested: 45% acceptance, 62% terminal utilisation.
- github.com/nauvalZulfikar/Mini-Digital-Twin

LLM-Generated Adaptive Shipper Decision Rules (*Solo builder, May 2026*)

- End-to-end system that generates literature-grounded crowd-shipper decision rules from traveller profiles, using RAG over a curated behavioural literature knowledge base (Punel & Stathopoulos 2017; Archetti et al. 2016; Le et al. 2019) + gpt-4o-mini. Direct implementation of methodology named verbatim in the Aston PhD posting; continuation of the LLM-for-supply-chain work in the MSc dissertation under Dr. Al-Bazi.
- github.com/nauvalZulfikar/LLM-Shipper-Profiles-

VRPOD MILP Optimisation, Vehicle Routing with Occasional Drivers (*Solo builder, May 2026*)

- MILP formulation of the foundational VRPOD problem (Archetti, Savelsbergh & Speranza, 2016), solved with `scipy.optimize.milp` (HiGHS). On a Birmingham-inspired toy instance (8 customers, 2 trucks, 3 crowd-shippers), integrating crowd-shipping **reduces total cost by 38.5%**.
- github.com/nauvalZulfikar/VRP-MILP

Crowd-Shipping ABM Behavioural Sweep (*Solo builder, May 2026*)

- 450-run parameter sweep (9 reward levels × 5 supply densities × 10 replications) with auto-generated findings linking results to literature. Supply effect: acceptance 45.34% (n=20) → 79.80% (n=100), a 34-percentage-point spread.
- github.com/nauvalZulfikar/Crowd-Shipping-ABM-

INDUSTRY PROJECTS

ML for Dam & Irrigation Gate Automation (Cibeureum) (2025) · ML feasibility analysis for infrastructure control system, proxy Digital Twin for hydraulic infrastructure.

Machine Scheduler Automation (Jan 2025) · Streamlit-based production scheduling with Gantt visualisation and real-time utilisation dashboards. Optimised production planning by 63% and minimised machine idle time.

CV-Job Description Suitability Checker (Nov 2024) · DeBERTa-v3 + RAG semantic matching that outperforms traditional ATS lexical filters. End-to-end ML pipeline with orchestration; codebase repurposed for the LLM Shipper Profiles research prototype above.

PUBLICATIONS

Zulfikar, N. (2025) *Performance Analysis in Sport Footwear Sales Prediction Using Machine Learning*. Comparative evaluation of multiple ML regressors for retail sales forecasting accuracy.

WORK EXPERIENCE

Public Works & Spatial Planning Department (PUTR), Bandung (Government · Data Scientist & Project Manager · Sep 2022 – Present) Automated ETL pipelines and built self-service dashboards across building permits, small-business, tourism, and spatial-planning data; embedded data-governance and privacy practice throughout. **45% faster project delivery, 20% data-efficiency gain.**

Syncwell, Birmingham, UK (Start-up · Marketing Data Scientist · Jul – Sep 2024) Predictive ML, A/B testing, automated campaign triggers, personalisation strategies. **40% CTR increase, 22% follower growth.**

PCOS Challenge, Birmingham, UK (Non-profit · Strategic Data Scientist Specialist · Jun – Aug 2024) Customer attribution model and automated report text extraction; interactive visualisations of patient financial concerns. **80% manual workload reduction.**

Bank Muamalat Indonesia, Jakarta (Corporation · Lead Data Scientist, Leadership Development Programme · Nov 2019 – Aug 2022) NLP models for credit portfolio analysis and risk scoring in a licensed banking context; A/B testing across marketing strategies; clustering for market segmentation; NLP-based app review analysis vs competitors. **+5.3% investment return.** Maintained model reliability through regular validation cycles, monitored model staleness, and ensured documentation compliance with regulatory and audit frameworks.

TECHNICAL SKILLS

Research & simulation methods: Agent-Based Modelling (mesa), Discrete-Event Simulation (simpy), Digital Twin architecture, Mixed-Integer Linear Programming (Pyomo, scipy.optimize.milp, HiGHS), Discrete choice modelling (MNL, Mixed Logit, Biogeme), AnyLogic (in progress), RAG over behavioural literature, Causal inference, Bayesian methods. **Languages & data:** Python, SQL, R, PySpark, NumPy, Scipy, pandas, scikit-learn, PyTorch, TensorFlow, Keras, FastAPI, Streamlit. **AI / ML:** Deep Learning, NLP, LLM (gpt-4o-mini, DeBERTa-v3), RAG, sentence-transformers, classification, regression, time-series forecasting, recommender systems, attribution modelling. **Governance & infrastructure:** Model validation, model reliability, model governance, audit-grade documentation, AWS, Docker, NoSQL, Hadoop, Git, ETL, workflow orchestration, Power BI, Tableau.

LANGUAGES

Indonesian (Native: reading, listening, writing, speaking) · **English** (Native, MSc English-medium: reading, listening, writing, speaking) · **Japanese** (Intermediate: reading, listening, writing, speaking)